

# Research on Intrinsic Motivation for Reinforcement Learning

## Correlation-Aware Reward Exploration (CARE)

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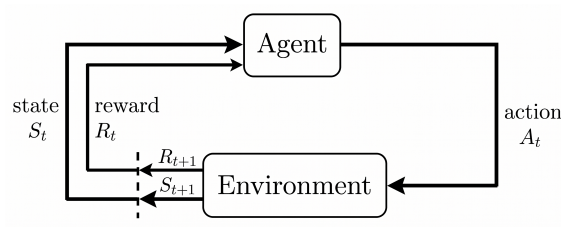
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# Background: Reinforcement Learning and Sparse Reward

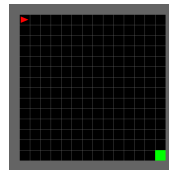


**Dense Reward**



MuJoCo Humanoid

**Sparse Reward**



MiniGrid Empty-16x16

## PPO as the Core RL Framework

- **Policy optimization:** Proximal Policy Optimization (PPO) is the base RL algorithm that updates the policy  $\pi_\theta$ .
- **Objective:** PPO maximizes the expected total *shaped* reward  $r_t$ , with its clipping mechanism keeping policy updates stable and effective.

Exploration methods that supply the intrinsic reward  $r_t^{int}$ :

### Novelty

*"Have I been here before?"*

**Count-based**

### Surprise

*"Did I predict that?"*

**ICM**

### Influence

*"Did I change the world?"*

**RIDE**

$$\text{Shaped reward: } r_t = r_t^{ext} + \beta r_t^{int}$$

PPO: Proximal Policy Optimization • ICM: Intrinsic Curiosity Module • RIDE: Rewarding Impact-Driven Exploration

## Fixed- $\beta$ Problem and Research Objective

- Same  $\beta$  for every state – but some states need more exploration than others.
- Same  $\beta$  throughout training – exploration need changes as the agent learns.
- Ignores the task – keeps exploring even when exploration does not help solve the task.

### Objective

*A method to adapt the intrinsic-reward coefficient automatically during training so that exploration better supports extrinsic-reward learning*

**Before:**  $r_t = r_t^{\text{ext}} + \beta r_t^{\text{int}}$

**CARE:**  $r_t = r_t^{\text{ext}} + \boxed{\beta_\psi(s_t)} r_t^{\text{int}}$

## Main idea

Raise intrinsic reward when exploration helps future extrinsic reward; reduce it when distracting.

- **Correlation** – align intrinsic reward with future extrinsic advantage.
- **Regularizer** – keep  $\beta_\psi$  near  $\beta_0$  in log-space.

$$\mathcal{L}_\beta(\psi) = \underbrace{-\mathbb{E}[\hat{I}_z \cdot \hat{A}_z^E]}_{\text{correlation}} + \underbrace{\lambda_{\text{reg}} (\log \beta_\psi(s) - \log \beta_0)^2}_{\text{log-space regularizer}}$$

# Experimental Design

|                   |                                                         |
|-------------------|---------------------------------------------------------|
| Base algorithm    | PPO                                                     |
| Intrinsic modules | Count-based, ICM, RIDE                                  |
| Compared methods  | PPO (no intrinsic) / PPO + fixed- $\beta$ / PPO + CARE  |
| Environments      | 6 MiniGrid sparse-reward tasks ( $10^6$ env steps each) |
| Metrics           | Success rate, episode return, learning speed, stability |
| Seeds             | {1, 2, 3, 4, 5}                                         |

## PPO Config

$K = 80$   
 $\epsilon_{\text{clip}} = 0.2$   
 $\gamma = 0.99$   
 $\lambda_{\text{GAE}} = 0.95$   
 $\text{lr}_{\text{actor}} = 3 \times 10^{-4}$   
 $\text{lr}_{\text{critic}} = 10^{-3}$   
rollout  $T = 4,000$

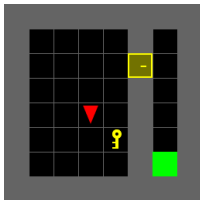
## CARE Config

$\beta_{\min} = 10^{-4}$   
 $\beta_{\max} = 5 \times 10^{-2}$   
 $\beta_0 = \sqrt{\beta_{\min} \beta_{\max}} \approx 2.24 \times 10^{-3}$   
 $\lambda_{\text{reg}} = 10^{-3}$   
 $\text{lr}_{\psi} = 5 \times 10^{-4}$   
weight decay =  $10^{-6}$   
grad-clip (L2) = 1.0

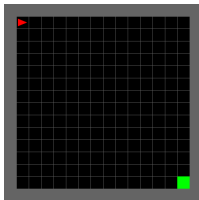
**Fixed- $\beta$  sweep:**  $\beta \in \{0.0005, 0.001, 0.005, 0.01, 0.05\}$ ; one  $\psi$ -step per PPO update

GAE: Generalized Advantage Estimation

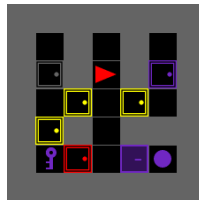
# Experimental Design



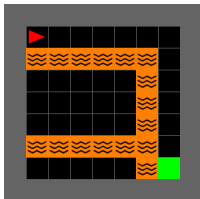
DoorKey-8x8



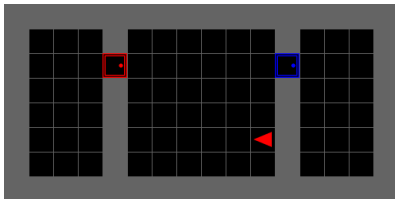
Empty-16x16



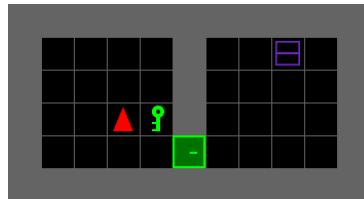
KeyCorridor-S3R3



LavaCrossing-S9N3



RedBlueDoors-8x8

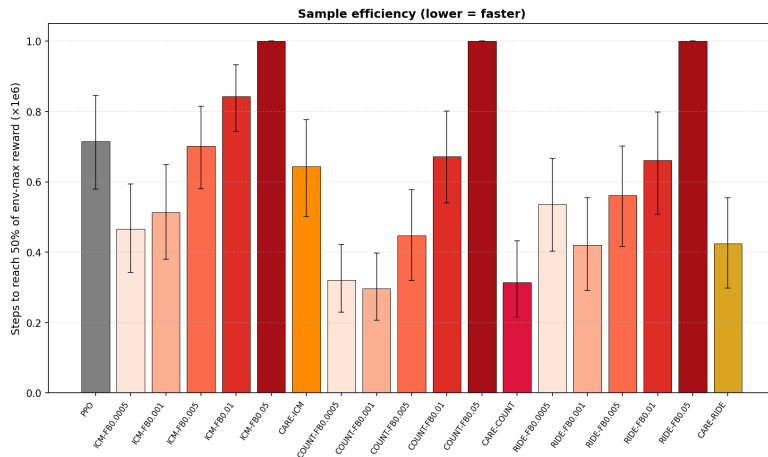


UnlockPickup

All sparse-reward, discrete grid. Red triangle = agent.

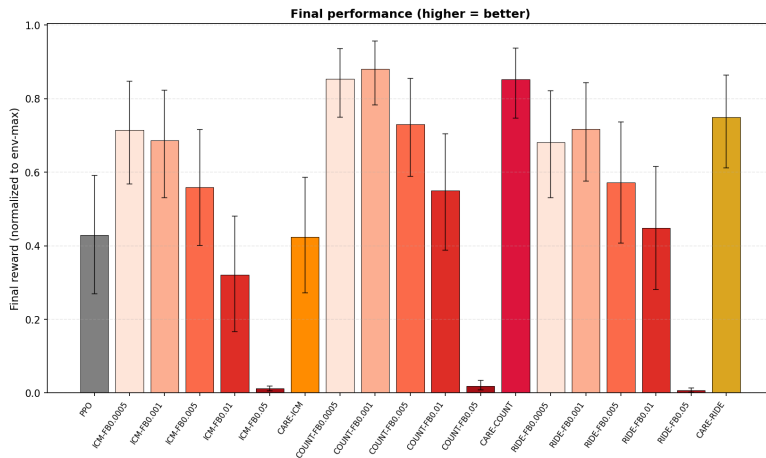


# Results – Sample Efficiency



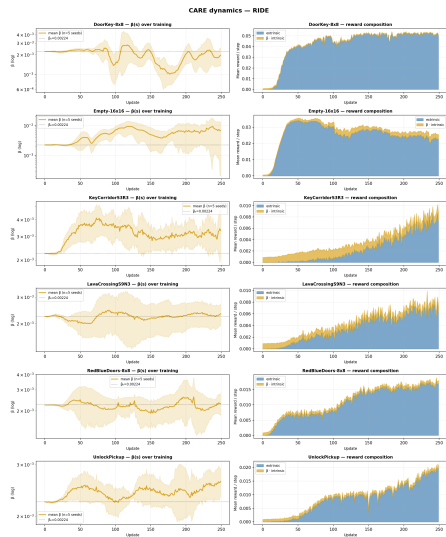
Lower is faster. Aggregated across 6 envs  $\times$  5 seeds (95% bootstrap confidence interval).

# Results – Final Performance



Higher is better. Final reward normalized to env-max, last 10 logs averaged.

# Analysis – Why CARE Works



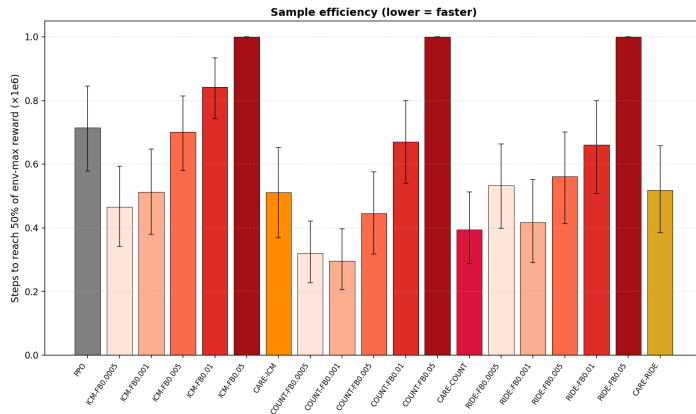
$\beta_{\psi}(s)$  (left) and reward composition (right) for 6 envs – CARE-RIDE.

- MiniGrid only (discrete, low-dim).
- Extra network + training objective.
- Sensitive to  $\beta_0$ ,  $\lambda_{\text{reg}}$ .
- Future: continuous control, image-based.

- Sparse rewards  $\Rightarrow$  hard exploration.
- Fixed  $\beta$  = one-size-fits-all.
- **CARE**: adaptive  $\beta_\psi(s)$  per state.

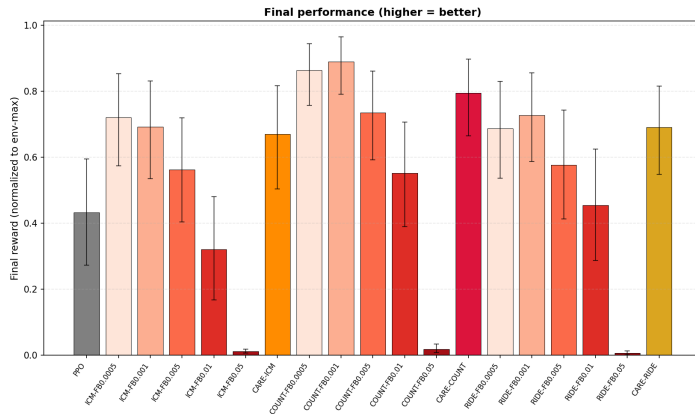
# Thank you for listening!

# [Backup] Extended $\beta$ Range – Sample Efficiency



$\beta$  sweep extended to  $\beta \in [10^{-8}, 1]$ .

# [Backup] Extended $\beta$ Range – Final Performance



$\beta$  sweep extended to  $\beta \in [10^{-8}, 1]$ .

## [Backup] CARE vs Best Fixed $\beta$ (all 18 cells)

| Env          | Module | CARE         | Best fixed | Best $\beta$ | Gap           |
|--------------|--------|--------------|------------|--------------|---------------|
| DoorKey-8x8  | COUNT  | 0.973        | 0.973      | 0.0005       | <b>+0.001</b> |
|              | ICM    | 0.770        | 0.973      | 0.0005       | -0.202        |
|              | RIDE   | 0.973        | 0.973      | 0.0005       | <b>+0.000</b> |
| Empty-16x16  | COUNT  | 0.959        | 0.975      | 0.0005       | -0.017        |
|              | ICM    | 0.575        | 0.975      | 0.0005       | -0.400        |
|              | RIDE   | 0.941        | 0.974      | 0.001        | -0.033        |
| KeyCorridor  | COUNT  | 0.831        | 0.828      | 0.001        | <b>+0.003</b> |
|              | ICM    | 0.000        | 0.457      | 0.001        | -0.457        |
|              | RIDE   | 0.548        | 0.703      | 0.001        | -0.155        |
| LavaCrossing | COUNT  | 0.372        | 0.530      | 0.001        | -0.158        |
|              | ICM    | 0.077        | 0.060      | 0.001        | <b>+0.017</b> |
|              | RIDE   | 0.543        | 0.512      | 0.0005       | <b>+0.031</b> |
| RedBlueDoors | COUNT  | 0.653        | 0.779      | 0.0005       | -0.126        |
|              | ICM    | 0.273        | 0.737      | 0.0005       | -0.464        |
|              | RIDE   | 0.593        | 0.794      | 0.001        | -0.201        |
| UnlockPickup | COUNT  | 0.936        | 0.936      | 0.0005       | <b>+0.000</b> |
|              | ICM    | 0.716        | 0.937      | 0.001        | -0.221        |
|              | RIDE   | <b>0.536</b> | 0.369      | 0.0005       | <b>+0.167</b> |